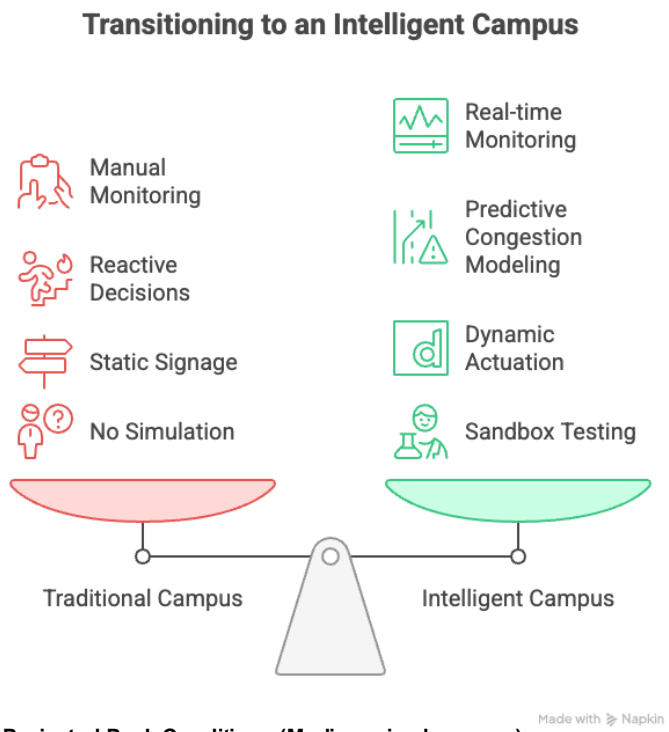
- ➔ Team name — **Nexus for AMD**
- ➔ Team leader name — **Likhith Kanigolla**
- ➔ Problem Statement — **AI for Smart Cities**
- ➔ Proposed Solution — Smart Campus Crowd Management using Digital Twin & IoT Sensors

Brief about the idea

- CrowdTwin is a real-time Digital Twin platform for university campus crowd management
- IoT sensors (cameras, occupancy sensors, road monitors) are deployed across the campus to capture live crowd data
- A 3D interactive map renders the campus with semantic building classification and real-time agent movement
- Three operational modes: Visualize (monitor), Actuate (control), Simulate (predict)
- AI-powered decision engine (Gemini/GPT) provides smart recommendations for crowd management
- Prevents stampedes, optimizes event logistics, and enables data-driven campus planning
- CrowdTwin observes, predicts, decides, and intervenes in real-time.”



Projected Peak Conditions (Medium-sized campus):
- 1,000–3,000 people moving simultaneously
- 8–15 high-density hotspots
- 2–5 min delay in manual response

Opportunities

CrowdTwin Expansion Opportunities

Emergency Response

Improve public safety and well-being

Multi-Campus Deployment

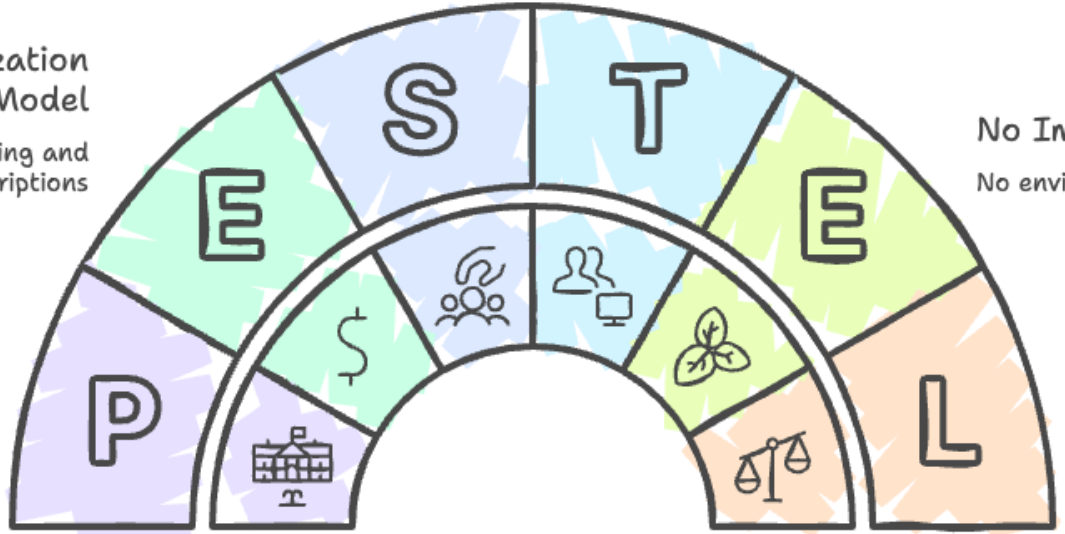
Scalable and integrated solutions

Monetization Model

Platform licensing and subscriptions

No Impact

No environmental impact



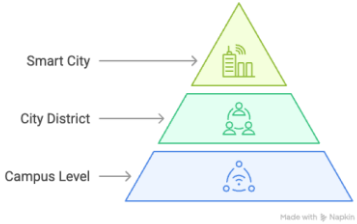
Smart City Integration

Government smart city initiatives

No Impact

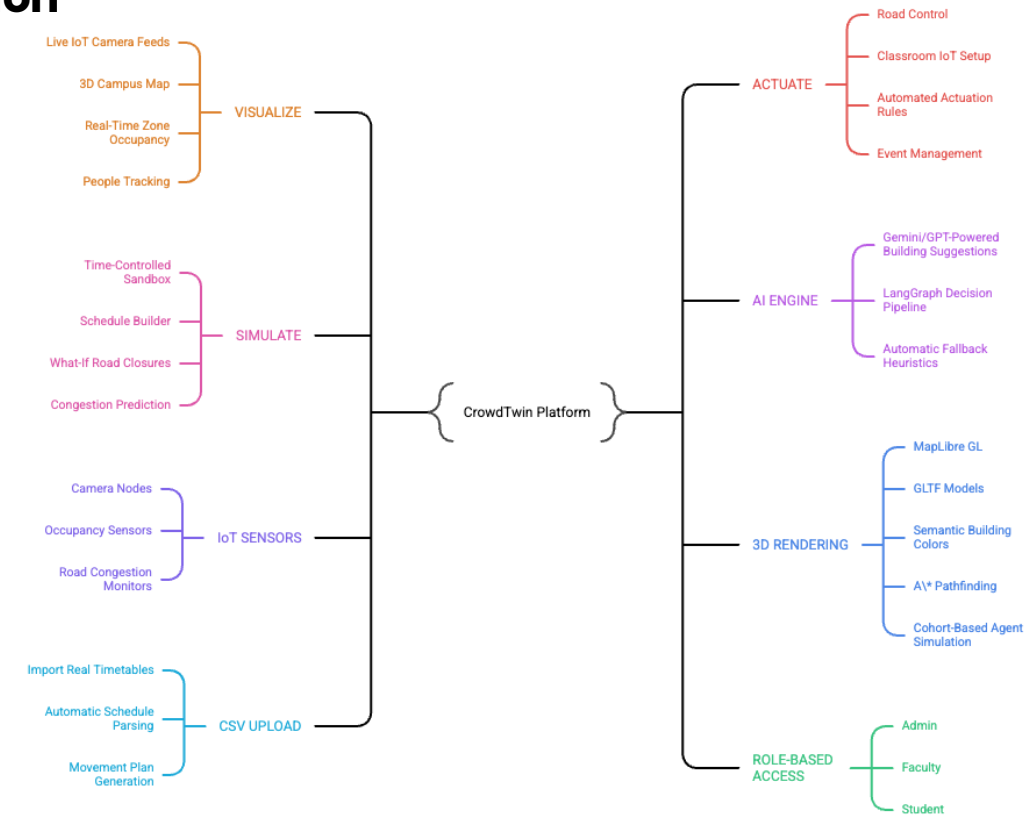
No legal impact

Urban Development Expansion Pyramid



List of features offered by the solution

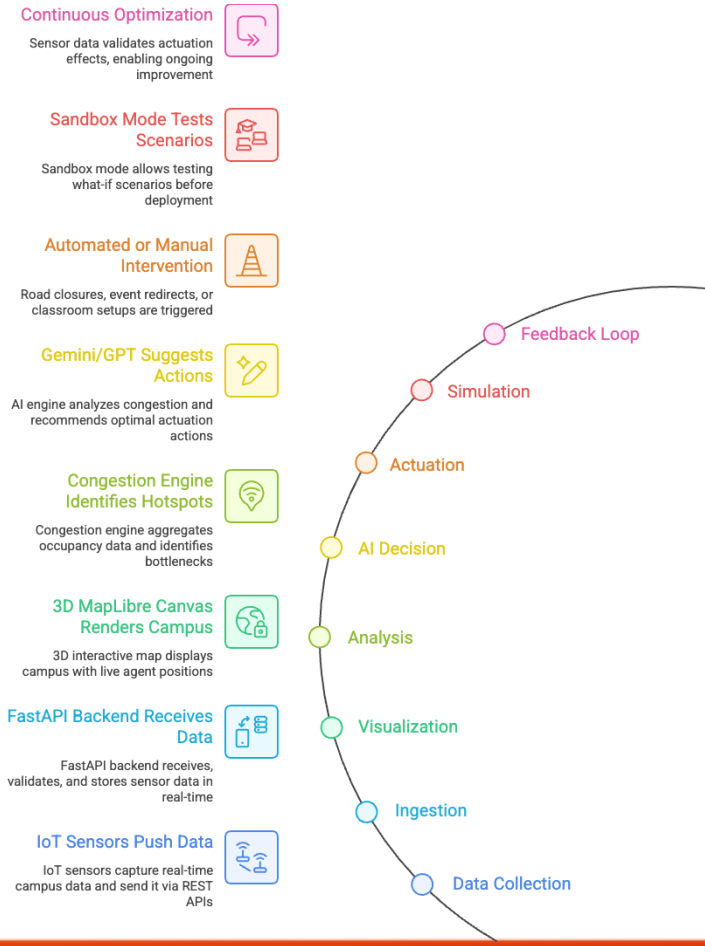
- **VISUALIZE:** Live IoT camera feeds, 3D campus map, real-time zone occupancy, people tracking
- **ACTUATE:** Road control (Open/Soft Close/Hard Close), classroom IoT setup, automated actuation rules, event management
- **SIMULATE:** Time-controlled sandbox (1x/5x/15x), schedule builder, what-if road closures, congestion prediction
- **AI ENGINE:** Gemini/GPT-powered building suggestions, LangGraph decision pipeline, automatic fallback heuristics
- **IoT SENSORS:** camera nodes at building entrances, occupancy sensors in buildings, road congestion monitors
- **3D RENDERING:** MapLibre GL with GLTF models, semantic building colors, A* pathfinding, cohort-based agent simulation
- **CSV UPLOAD:** Import real timetables, automatic schedule parsing, movement plan generation
- **ROLE-BASED ACCESS:** Admin (full control), Faculty (classroom setup), Student (limited actuation)



Process flow diagram or Use-case diagram

Process Flow:
IoT → Digital Twin → AI → Actuation

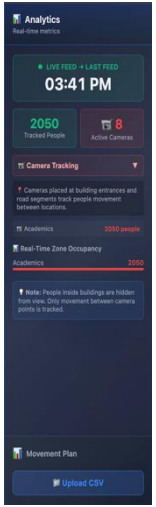
- DATA COLLECTION: IoT sensors (cameras, occupancy, road monitors) push data via REST APIs
- INGESTION: FastAPI backend receives, validates, and stores sensor data in real-time
- VISUALIZATION: 3D MapLibre canvas renders campus with live agent positions from sensor data
- ANALYSIS: Congestion engine aggregates occupancy data, identifies hotspots and bottlenecks
- AI DECISION: Gemini/GPT analyzes congestion patterns and suggests optimal actuation actions
- ACTUATION: Road closures, event redirects, and classroom setup triggered automatically or manually
- SIMULATION: Sandbox mode allows testing what-if scenarios before deploying changes
- FEEDBACK LOOP: Sensor data validates actuation effects, enabling continuous optimization



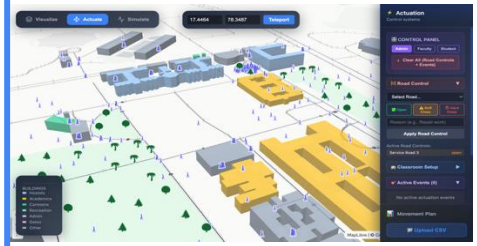
Wireframes/Mock diagrams of the proposed solution

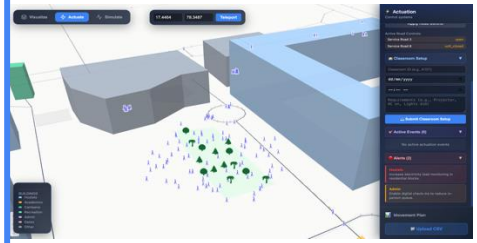
See attached screenshots: Visualize Mode, Actuate Mode, Simulate Mode, 3D Map View

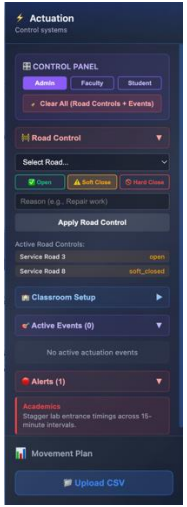
Visualisation



Actuation(Control)







Simulation

Architecture diagram of the proposed solution

System Architecture

FRONTEND LAYER: React 18 + Vite 5 + MapLibre GL JS (3D rendering)

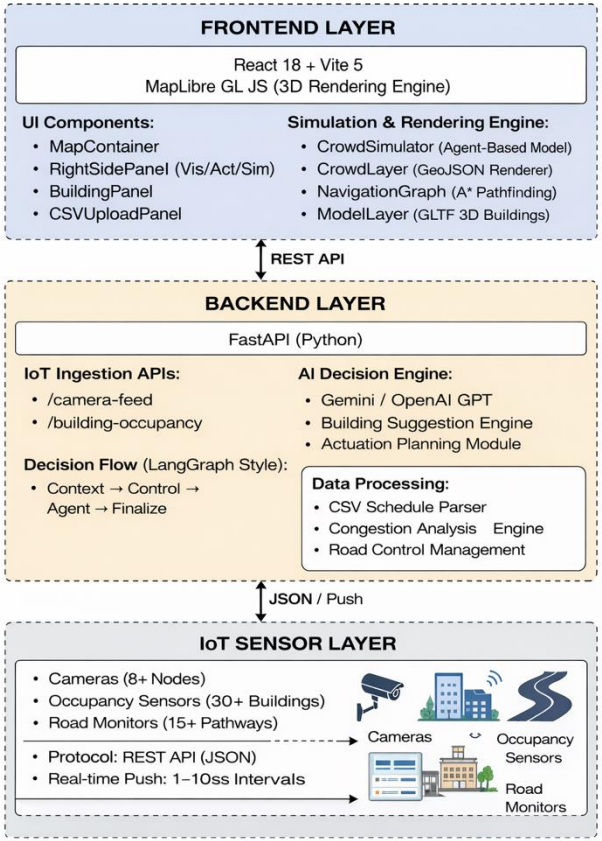
- → Components: MapContainer, RightSidePanel (Vis/Act/Sim), BuildingPanel, CSVUploadPanel
- → Engine: CrowdSimulator (agents), CrowdLayer (GeoJSON), NavigationGraph (A* pathfinding), ModelLayer (GLTF)

BACKEND LAYER: FastAPI (Python) — REST API server

- → IoT Ingestion: /camera-feed, /building-occupancy (real-time sensor data)
- → AI Engine: Gemini / OpenAI GPT — building suggestions, actuation planning
- → Actuation: LangGraph-style decision graph (Context → Control → Agent → Finalize)
- → Data: CSV schedule parser, congestion analysis, road control management

IoT SENSOR LAYER: Cameras (8+ nodes), Occupancy Sensors (30+ buildings), Road Monitors (15+ pathways)

- → Protocol: REST API (JSON), Real-time push (1-10s intervals)



Technologies to be used in the solution

- FRONTEND: React 18, Vite 5, MapLibre GL JS 4.x, GLTF/GLB 3D Models
- BACKEND: FastAPI (Python 3.10+), Pydantic, Uvicorn
- AI/ML: Google Gemini API, OpenAI GPT API, LangGraph-style decision pipeline
- 3D ENGINE: MapLibre GL JS with custom WebGL layers, A* pathfinding, GeoJSON rendering
- IoT PROTOCOLS: REST API (JSON), Real-time sensor data ingestion
- DATA FORMATS: GeoJSON (spatial), CSV (schedules), JSON (API communication)
- MAPPING: OpenStreetMap data, OSM building footprints, campus pathways
- DEPLOYMENT: Node.js 18+, Python 3.10+, Vite dev server + Uvicorn

	Vite	MapLibre GL JS	gltTF 3D Models
FastAPI	Pydantic	uvicorn Uvicorn	Google Gemini Google
OpenAI GPT	LangGraph	MapLibre GL JS 4.x MapLibre GL JS	Pathfinding
REST API	GeoJSON	CSV	JSON
OpenStreetMap	node js / Vite dev server +	python / Vite + Uvicorn	Vite uvicorn

Usage of AMD Products/Solutions(Proposed)

- AMD GPU Acceleration: WebGL-powered 3D rendering of 1200+ agents in real-time on MapLibre GL
- AMD Ryzen Processors: Backend AI inference and congestion analysis runs efficiently on AMD hardware
- AMD Instinct (Future): Scale AI decision pipeline to handle campus-wide sensor data in real-time
- AMD ROCm (Future): Optimize crowd prediction models using AMD's open-source ML framework
- AMD Embedded (IoT): IoT edge processing on AMD Xilinx FPGAs for camera-based people counting
- Performance: Smooth 60fps 3D rendering with 1000+ animated agents on AMD Radeon GPUs

Estimated implementation cost (TBD)

- IoT Cameras, Occupancy Sensors,
- Edge Computing Hardware (AMD Xilinx FPGAs)
- Cloud Server (FastAPI + AI APIs)
- AI API Costs (Gemini/GPT
- Development & Maintenance: Open Source

Currently using free tier deployments, if support given services/compute from AMD can be used to deploy with in the Smart City Living Lab –IIITH Campus

Prototype Assets (V1)

➔ GitHub Public Repository Link
https://github.com/likhithkanigolla/CrowdTwin

➔ Demo Video Link (Max: 3 Minutes)
https://drive.google.com/drive/folders/14enyEwmFWi9_9jwwZTjx1caFQ9Bj-fJf?usp=sharing

Real Time Validation

IoT Sensor Deployment Plan(at Smart City Living Lab)

- Camera Nodes (8+): Deployed at all building entrances and main road intersections
- → Data: People count, direction (in/out), GPS coordinates, timestamps
- → Frequency: Real-time (1-5 second intervals)

- Building Occupancy Sensors (30+): Installed in all classified buildings
- → Data: Current occupancy, capacity, last update timestamp
- → Categories: Hostels, Academics, Canteens, Recreation, Admin, Gates

- Road Monitors (10+): Placed along campus pathways
- → Data: Traffic flow, congestion level, road status
- → Control: Open / Soft Close / Hard Close via REST API

- Environmental Sensors (Future): Temperature, humidity, air quality in high-density zones

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S

Thank you!